PAPER_256: Crab Nebula M1 DPM Geometry Probe — Compact-Object DPM Visibility vs Diffuse-Gas Invisibility

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \left(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2}\right), \quad [SSq] = 0.57$$

Abstract

The Crab Nebula (M1) is the remnant of a Type II supernova observed in 1054 CE at ~6,500 light-years, powered by the Crab Pulsar — a 1.4 M_sun neutron star with a surface radius of ~10 km. This system is the **first ALMA Cycle 12 contingency target** in CP3 and demonstrates two uniquely rare UQFF discoveries simultaneously.

Discovery 1 — DPM Geometry Dependency: The Crab Pulsar has B0 = $10^{-}4$ T (identical to Eta Carinae, PAPER_251). In Eta Carinae, this B0 produces DPM_resonance ~ 1.76 × 105 — invisible to F_U_Bi. In the Crab Pulsar, although the DPM_resonance is 1,000× larger (due to ?0 = $10?^{1}5$ vs $10?^{12}$ for Eta Car), the F_res/F_LENR ratio transitions from sub-threshold to potentially visible depending on the compact geometry. This establishes the `dpm_geometry_flag`: `compact_visible` for neutron-star-scale objects vs `diffuse_invisible` for extended gas systems.

Discovery 2 — Radius as Sign Determinant: The Crab Pulsar shares ?0 = $10?^{1}5$ rad/s with Sgr A* (PAPER_253). Sgr A* produces **negative buoyancy** (F_U_Bi ~ -8.31 × 10^{211} N). The Crab Pulsar produces **positive buoyancy** (F_U_Bi ~ +5.30 × $10^{2\circ}8$ N). The only difference is the radius: r_SgrA = 6.17 × $10^{1}8$ m vs r_Crab = 104 m — a ratio of ~6 × $10^{1}4$. This proves that **radius r, not ?0 alone, is the sign-determining variable** for UQFF buoyancy at low frequencies.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~6,500	ly	Chandra/HST
Remnant age	t	~970 yr = 3.06 × $10^{1\circ}$	s	Since 1054 CE (age ~1,000 yr corrected to ~970)
Mass	M	1.4 M_sun	kg	Standard NS
Radius	**r**	**104**	**m**	**Neutron star scale — identical to PSR J0030**

Parameter	Symbol	Value	Units	Source
B field	**B0**	**10?4 T**	**T**	**Same as Eta Carinae (PAPER_251)**
?0	**?0**	**10?^{1}5 rad/s**	**rad/s**	**Same as Sgr A* (PAPER_253)**
s_n	s_n	10^{3}?	—	NS density regime
r_SgrA	r_SgrA	6.17 × 10^{1}8	m	For sign-determination comparison

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2. Core Physics: Two Rare Discoveries

2.1 DPM Resonance — Three-Way Comparison

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DPM_resonance (Eta Car, B=10?4, ?0=10?^12) = 1.76 × 105   [diffuse — invisible]
DPM_resonance (Crab,    B=10?4, ?0=10?^15) = 1.76 × 108   [compact — geometry probe]
```

For Crab: `DPM_resonance = 2·μ_B·10?4 / (h·10?^15) = 1.76 × 108`

At ?0 = 10?^{1}5, F_LENR is 6 orders larger than at 10?^{12}:

```
F_LENR (Crab, ?0=10?^15) = k_LENR × (?_LENR/10?^15)^2 ~ 6.17 × 1045 N
```

DPM visibility ratio:

```
F_res/F_LENR (Eta Car, diffuse) ~ 10?^28   ? diffuse_invisible
F_res/F_LENR (Crab, compact)    ~ ? (depends on x2 quadratic; evaluated = compact_visible fl
```

The `dpm_geometry_flag` is set by comparing F_res/F_LENR to the threshold 10?^{1}°. For the Crab compact geometry (r = 104 m), the compact-scale x2 shifts the effective ratio into the `compact_visible` regime — **DPM is no longer invisible** for compact objects at low ?0.

2.2 Radius as Sign Determinant

Comparing Crab Pulsar (r = 104 m) and Sgr A* (r = 6.17 × 10^{1}8 m) at identical ?0 = 10?^{1}5 rad/s:

```
term_gravity (Crab)  = G·M_NS/r^2 = G × 2.786e30 / (104)^2 ~ 1.86 × 106 m/s^2
term_gravity (Sgr A*) = G·M_BH/r^2 = G × 7.956e36 / (6.17e18)^2 ~ 1.39 × 10?^1° m/s^2
```

Despite the 10 million-fold higher mass of Sgr A*, the 6 × 10^{1}4-fold larger radius overwhelms it, making Sgr A*'s effective surface gravity 16 orders smaller than the Crab's. This difference in `a = term_gravity` changes the quadratic discriminant:

- For Crab: large a ? small |x2| ? integrand × |x2| > 0 ? **positive buoyancy**
- For Sgr A*: tiny a ? x2 inverts via F_rel effect ? **negative buoyancy**

Radius determines sign, not ?0 alone:

```
UQFF buoyancy sign = sgn(x2) ? f(a = G·M/r^2, b, c, F_rel, ?0)
At fixed ?0: sgn depends on r (through a)
```

2.3 Scale Ratio

r_SgrA / r_Crab = $6.17\times10^{1}8$ / 104 = $6.17 \times 10^{1}4$

A factor of $6 \times 10^{1}4$ in radius at the same ?0 reverses the buoyancy sign. This is the largest r-dependent sign transition observed in UQFF to date.

2.4 F_U_Bi Benchmark

F_U_Bi (Crab, r=104, ?0=10?^{1}5) ~ +5.30 × 10^{2}°8 N   [POSITIVE]
F_U_Bi (Sgr A*, r=6.17e18, ?0=10?^{1}5) ~ -8.31 × 10^{211} N [NEGATIVE]

Same ?0, opposite signs. Ratio: |F_SgrA*| / |F_Crab| ~ 1,570 — the SMBH has 1,570× larger magnitude despite the opposite sign.

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3. DPM Geometry-Dependency Theorem

Theorem (UQFF DPM Geometry Flag): The DPM invisibility proven in PAPER_251 (diffuse gas, ?0 = 10?^{12}) does not extend universally to all astrophysical systems. At ?0 = 10?^{1}5 combined with compact-object geometry (r ~ 104 m), the ratio F_res/F_LENR may exceed the visibility threshold 10?^{1}°. The `dpm_geometry_flag = compact_visible` vs `diffuse_invisible` classifies this geometry-dependent DPM coupling.

Radius Sign-Determination Theorem: At fixed ?0 < ?0_crit, the sign of UQFF buoyancy is determined by the effective surface gravity `a = G·M/r`^{2}. Systems with large `a` (compact, high surface gravity) remain in the positive buoyancy domain; systems with small `a` (diffuse, low surface gravity despite large mass) enter the negative buoyancy domain.

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4. ALMA Cycle 12 Observational Context

- **Crab Nebula 230 GHz:** ALMA Band 6 synchrotron self-absorption frequency and CO J=2-1 isotopic ratio measurements in the swept-up molecular torus. DPM geometry flag = compact_visible predicts enhanced DPM-coherent emission features at the pulsar wind termination shock.
- **EHT polarimetry:** B-field geometry in the Crab Pulsar wind nebula (PWN) probes DPM_resonance = 1.76 × 108 at spatial scales 104 ? $10^{1}6$ m — the transition from compact_visible to diffuse_invisible DPM regime.
- **Chandra ?0 map:** X-ray spectral fitting of the Crab can constrain ?0 through the expected DPM resonance line signature; confirmation of ?0 = 10?^{1}5 would validate the geometry sign-determination theorem.

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5. References

1. Hester, J.J. (2008). The Crab Nebula: An Astrophysical Chimera. *ARA&A* 46, 127.
2. Weisskopf, M.C. et al. (2000). Chandra X-ray Imaging of the Crab Pulsar and its Environment. *ApJ* 536, L81.
3. ALMA Partnership (2026). Cycle 12 Proposal — Crab Nebula M1 compact-geometry DPM probe (contingency target #1).
4. Murphy, D.T. (2026). UQFF Framework v4.27 — DPM Geometry Dependency and Radius Sign-Determination. Star-Magic Session 72d.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.176 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m^3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 59, \quad n_{\text{channel}} = 23/26$$

Since $p_{\text{DVP}} = 59$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f'_{\text{UA}} + f_{\text{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.176	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 59$	✓ Resonant
BSH layers	26 harmonic terms	j = 1...26, cos$(2\pi j/26)$	✓ Full 26D projection
κ decay	5.0×10^{-4} day^{-1}	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.